

A Technical Introduction to Deep Q-network

Section 1

Max pooling In 1990 Yamaguchi et al. introduced the concept of max pooling, a fixed filtering operation that calculates and propagates the maximum value of a given region. They did so by combining TDNNs with max pooling to realize a speaker-independent isolated word recognition system. In their system they used several TDNNs per word, one for each syllable. The results of each TDNN over the input signal were combined using max pooling and the outputs of the pooling layers were then passed on to networks performing the actual word classification. In a variant of the neocognitron called the cresceptron, instead of using Fukushima's spatial averaging with inhibition and saturation, J. Weng et al. in 1993 used max pooling, where a downsampling unit computes the maximum of the activations of the units in its patch, introducing this method into the vision field. Max pooling is often used in modern CNNs.

Section 2

Number of filters Since feature map size decreases with depth, layers near the input layer tend to have fewer filters while higher layers can have more. To equalize computation at each layer, the product of feature values v with pixel position is kept roughly constant across layers. Preserving more information about the input would require keeping the total number of activations (number of feature maps times number of pixel positions) non-decreasing from one layer to the next. The number of feature maps directly controls the capacity and depends on the number of available examples and task complexity.

Section 3

Convolutional layers In a CNN, the input is a tensor with shape: (number of inputs) $\times$ (input height) $\times$ (input width) $\times$ (input channels). After passing through a convolutional layer, the image becomes abstracted to a feature map, also called an activation map, with shape: (number of inputs) $\times$ (feature map height) $\times$ (feature map width) $\times$ (feature map channels). Convolutional layers convolve the input and pass its result to the next layer. This is similar to the response of a neuron in the visual cortex to a specific stimulus. Each convolutional neuron processes data only for its receptive field.

Section 4

Cultural heritage and 3D-datasets As archaeological findings such as clay tablets with cuneiform writing are increasingly acquired using 3D scanners, benchmark datasets are becoming available, including HeiCuBeDa providing almost 2000 normalized 2-D and 3-D datasets

prepared with the GigaMesh Software Framework. So curvature-based measures are used in conjunction with geometric neural networks (GNNs), e.g. for period classification of those clay tablets being among the oldest documents of human history.

Section 5

Max norm constraints Another form of regularization is to enforce an absolute upper bound on the magnitude of the weight vector for every neuron and use projected gradient descent to enforce the constraint. In practice, this corresponds to performing the parameter update as normal, and then enforcing the constraint by clamping the weight vector $w \rightarrow \{\vec{w}\}$ of every neuron to satisfy $\|w\|_2 < c$. Typical values of c are order of 3–4. Some papers report improvements when using this form of regularization.

Section 6

Convolutional deep belief networks (CDBN) have structure very similar to convolutional neural networks and are trained similarly to deep belief networks. Therefore, they exploit the 2D structure of images, like CNNs do, and make use of pre-training like deep belief networks. They provide a generic structure that can be used in many image and signal processing tasks. Benchmark results on standard image datasets like CIFAR have been obtained using CDBNs.

Section 7

Animal behavior detection CNNs have been applied in ecological and behavioral research to automatically detect and quantify animal behavior from visual data, enabling identification of animals, tracking of individuals, estimation of pose, and classification of specific actions such as feeding, and social interactions. Combined with multi-object tracking and temporal modeling, these systems can extract behavioral sequences over extended recordings, reducing reliance on manual annotation and increasing throughput for studies of individual variation, social networks, and collective dynamics.

Section 8

simple cells, whose output is maximized by straight edges having particular orientations within their receptive field complex cells, which have larger receptive fields, whose output is insensitive to the exact position of the edges in the field. Hubel and Wiesel also proposed a cascading model of these two types of cells for use in pattern recognition tasks.

Section 9

Dilation Dilation involves ignoring pixels within a kernel. This reduces processing memory potentially without

significant signal loss. A dilation of 2 on a 3x3 kernel expands the kernel to 5x5, while still processing 9 (evenly spaced) pixels. Specifically, the processed pixels after the dilation are the cells (1,1), (1,3), (1,5), (3,1), (3,3), (3,5), (5,1), (5,3), (5,5), where (i,j) denotes the cell of the i-th row and j-th column in the expanded 5x5 kernel. Accordingly, dilation of 4 expands the kernel to 7x7.

Section 10

Natural language processing CNNs have also been explored for natural language processing. CNN models are effective for various NLP problems and achieved excellent results in semantic parsing, search query retrieval, sentence modeling, classification, prediction and other traditional NLP tasks. Compared to traditional language processing methods such as recurrent neural networks, CNNs can represent different contextual realities of language that do not rely on a series-sequence assumption, while RNNs are better suitable when classical time series modeling is required.

Section 11

Number of parameters Another simple way to prevent overfitting is to limit the number of parameters, typically by limiting the number of hidden units in each layer or limiting network depth. For convolutional networks, the filter size also affects the number of parameters. Limiting the number of parameters restricts the predictive power of the network directly, reducing the complexity of the function that it can perform on the data, and thus limits the amount of overfitting. This is equivalent to a "zero norm".

Section 12

Human interpretable explanations End-to-end training and prediction are common practice in computer vision. However, human interpretable explanations are required for critical systems such as self-driving cars. With recent advances in visual salience, spatial attention, and temporal attention, the most critical spatial regions/temporal instants could be visualized to justify the CNN predictions.